1D Seismic Trace Visualizer

Computational Geophysics & Forward Modeling

Ved Dipakkumar Parmar
B.Tech Petroleum Engineering
Pandit Deendayal Energy University (PDEU)

Documentation Coordinator
SEG × SPG × EAGE Student Chapter

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1. Executive Summary

This report distills the complex mathematical foundations of seismic exploration into a streamlined, deployable data science project. By modeling acoustic impedance, reflection coefficients, and convolutional mathematics using Python, this visualizer simulates the exact processes utilized by major energy conglomerates for subsurface reservoir characterization.

2. Seismic Exploration

Exploration geophysics utilizes the physical properties of the Earth to map structural traps and potential fluid distributions. This project relies on **forward modeling**—starting with a known geological state (a rock column) and mathematically simulating the resulting seismic response.

> ### Why Forward Modeling?
>
> It allows petroleum engineers to test hypotheses. If a well log suggests a specific sequence of shale and sandstone, forward modeling dictates exactly what the resulting seismic trace *should* look like.

3. Acoustic Wave Propagation

When a seismic source (like a vibroseis truck) discharges energy, compressional waves (P-waves) travel downward. Their velocity depends entirely on the rock matrix and the fluids filling its pore spaces. Dense rocks transmit energy quickly, while porous, fluid-filled rocks slow the wave down.

*Note: We do not directly image the rocks themselves; we image the **changes** in the rock's physical properties.*

4. Petrophysics & Boundaries

Acoustic Impedance (Z) is the product of the rock's bulk density (ρ) and the compressional wave velocity (V_p).

$$Z = \rho \times V_p$$

When an acoustic wave encounters a boundary between two different rock impedances, a portion of the energy reflects back. This ratio is the **Reflection Coefficient (RC)**:

$$RC = (Z_2 - Z_1) / (Z_2 + Z_1)$$

- **Positive Peak (+):** Moving from soft rock to hard rock (e.g., shale to limestone caprock).
- **Negative Trough (−):** Moving from hard rock to porous, oil-bearing sandstone.

5. Mathematical Architecture

To render a seismic trace on a computer, geological strata are converted into a discrete mathematical array. We use a transient pulse called the **Ricker Wavelet**, defined as the negative normalized second derivative of a Gaussian function.

$$w(t) = (1 - 2\pi^2 f^2 t^2)\, e^{-\pi^2 f^2 t^2}$$

A recorded seismic trace **S(t)** is mathematically generated by convolving the earth's reflectivity series **R(t)** with the seismic source wavelet **w(t)**.

6. Python Implementation

The code utilizes NumPy for vectorized mathematics and Matplotlib for standard visualization, mirroring modern enterprise data environments.

```
import numpy as np
import matplotlib.pyplot as plt

# 1. Create the Soundwave (Ricker Wavelet)
frequency = 30.0  # Dominant frequency in Hz
time = np.arange(-0.05, 0.05, 0.002)
pft2 = (np.pi * frequency * time)**2
wavelet = (1 - 2 * pft2) * np.exp(-pft2)

# 2. Simulate Earth Reflectivity (RC)
rc = np.zeros(250)
rc[50] = 0.5   # Hard limestone caprock
rc[100] = -0.4 # Porous sandstone (hydrocarbon reservoir)
rc[180] = 0.3  # Hard basement layer

# 3. Convolution Engine
trace = np.convolve(rc, wavelet, mode='same')
```

7. Interpretation & Application

The visual peaks and troughs correlate directly to actual subsurface rock properties:

- **Depth 50 (RC = +0.5):** Strong positive peak indicating a transition to high-impedance rock, acting as an impermeable seal.
- **Depth 100 (RC = -0.4):** Pronounced negative trough. This is the hallmark of a **reservoir pay zone**. The drop in impedance is caused by increased porosity and the presence of fluids.

Enterprise Workflow Integration

This Python architecture forms the backbone of large-scale exploration software utilized by companies like Reliance Industries. Understanding forward-modeling programmatically equips engineers to perform AVO (Amplitude Versus Offset) analysis and identify drilling hazards before field operations ever commence.